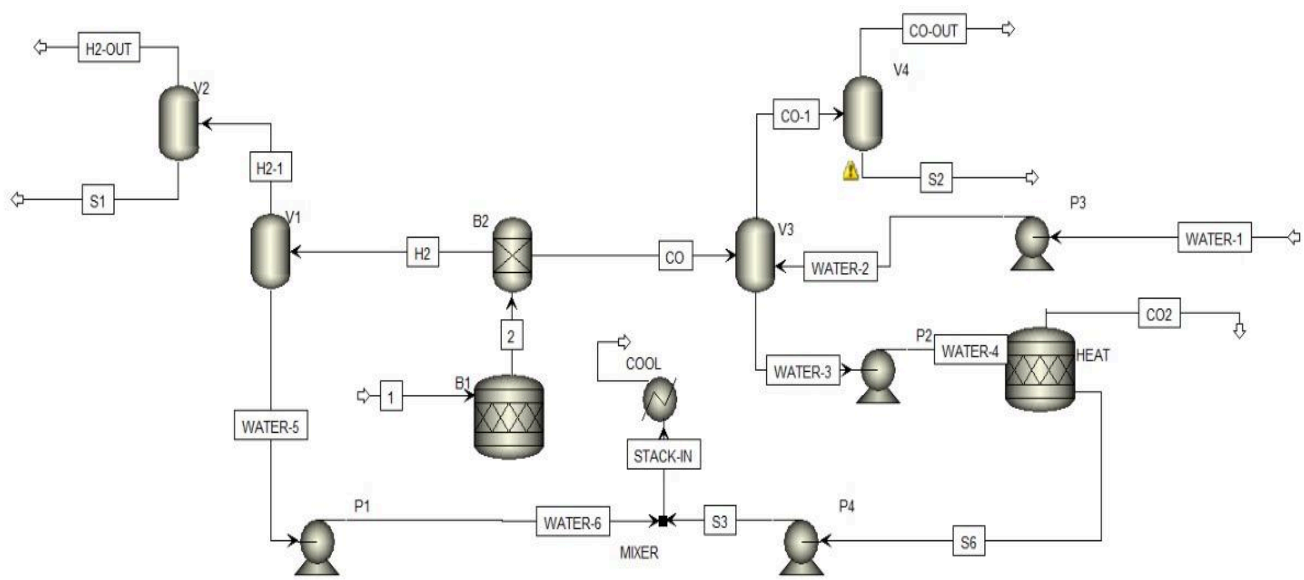
Title: Syngas/hydrogen production from self-supporting carbon sacrificial anode-assisted water electrolysis

Aim: To find the gas composition in output stream of H2 and CO.

Procedure:

1. New Aspen file was created.
2. Components were added: C, CO2, CO, O2, H2, KOH, KHCO3, H2O.
3. Elec Wizard was used for H2O and KOH to form H+, K+, OH-.
4. Base Method was selected as ELECNRTL.
5. Model was simulated.
6. From the given data table values of each component was entered.
7. Simulation was run and results were obtained.

Block Diagram



Result:

H2 Stream result

− Mole Flows	**kmol/hr**	**47.8286**
CO	kmol/hr	0
C	kmol/hr	0
H2O	kmol/hr	3.24724
H2	kmol/hr	25.9779
O2	kmol/hr	12.989
KOH	kmol/hr	0
CO2	kmol/hr	0
H+	kmol/hr	0
K+	kmol/hr	2.8072
OH-	kmol/hr	2.8072
K2CO3(S)	kmol/hr	0
KOH(S)	kmol/hr	0
KHCO3(S)	kmol/hr	0
HCO3-	kmol/hr	0
CO3--	kmol/hr	0

CO Stream Result

− Mole Flows	**kmol/hr**	**8.86165**
CO	kmol/hr	0
C	kmol/hr	0
H2O	kmol/hr	3.24724
H2	kmol/hr	0
O2	kmol/hr	0
KOH	kmol/hr	0
CO2	kmol/hr	0
H+	kmol/hr	0
K+	kmol/hr	2.8072
OH-	kmol/hr	2.8072
K2CO3(S)	kmol/hr	0
KOH(S)	kmol/hr	0
KHCO3(S)	kmol/hr	0
HCO3-	kmol/hr	0
CO3--	kmol/hr	0